

Cyclic Extensions

Recall: The norm on F/K is defined as

$$N_{F/K}(\alpha) = \prod \sigma(\alpha)$$

and we define the trace on F/K as

$$\text{Tr}_{F/K}(\alpha) = \sum \sigma(\alpha)$$

where both are distinct embeddings $F \hookrightarrow \bar{K}$ fixing K pointwise.

Theorem: Let F/K be finite and separable.

(i) $N_{F/K}(\alpha\beta) = N_{F/K}(\alpha) N_{F/K}(\beta)$

(ii) $\text{Tr}_{F/K}(\alpha + \beta) = \text{Tr}_{F/K}(\alpha) + \text{Tr}_{F/K}(\beta)$

(iii) $\text{Tr}_{F/K}(k\alpha) = k \text{Tr}_{F/K}(\alpha)$

(iv) For all $\alpha \in K$, $N_{F/K}(\alpha) = \alpha^{[F:K]}$

(v) For all $\alpha \in K$, $\text{Tr}_{F/K}(\alpha) = [F:K]\alpha$.

(vi) $N_{F/K}(\alpha), \text{Tr}_{F/K}(\alpha) \in K$. In particular, if $f(x)$ is the minimal polynomial of α over K , then

$$f(x)^{[F:K(\alpha)]} = x^{[F:K(\alpha)]} - [F:K(\alpha)] \text{Tr}_{F/K}(\alpha) x^{[F:K]-1} \\ + \dots + (-1)^{[F:K]} N_{F/K}(\alpha)^{[F:K(\alpha)]}$$

(vii) If E is an intermediate field

$$N_{E/K}(N_{F/E}(\alpha)) = N_{F/K}(\alpha)$$

$$\text{Tr}_{E/K}(\text{Tr}_{F/E}(\alpha)) = \text{Tr}_{F/K}(\alpha)$$

Proof: (i) $N_{F/K}(\alpha\beta) = \prod_{\sigma: F \rightarrow \bar{K}} \sigma(\alpha\beta) = \prod \sigma(\alpha) \prod \sigma(\beta) \\ = N_{F/K}(\alpha) N_{F/K}(\beta).$

$$(ii) \operatorname{Tr}(\alpha + \beta) = \sum \sigma(\alpha) + \sum \sigma(\beta) = \operatorname{Tr}(\alpha) + \operatorname{Tr}(\beta).$$

(iv) If $\alpha \in K$, $\sigma(\alpha) = \alpha \quad \forall \sigma$, so

$$N(\alpha) = \prod \sigma(\alpha) = \alpha^{[F:K]}$$

(vi) If $F = K(\alpha)$, then the roots of the minimal polynomial of α over K are exactly the $\sigma(\alpha)$, so

$$f(x) = \prod_{\sigma: F \rightarrow \bar{K}} (x - \sigma(\alpha)) = x^n - \sum \sigma(\alpha) x^{n-1} + \dots + (-1)^n \prod \sigma(\alpha)$$

where $n = \deg(f) = [F:K]$. In general, assuming (vii)

$$\begin{aligned} N_{F/K}(\alpha) &= N_{K(\alpha)/K}(N_{F/K(\alpha)}(\alpha)) \\ &= N_{K(\alpha)/K}(\alpha^{[F:K(\alpha)]}) \\ &= N_{K(\alpha)/K}(\alpha)^{[F:K(\alpha)]} \end{aligned}$$

Similarly, $\operatorname{Tr}_{F/K}(\alpha) = [F:K(\alpha)] \operatorname{Tr}_{K(\alpha)/K}(\alpha)$, so

$$f(x) = x^{[K(\alpha):K]} - \operatorname{Tr}_{K(\alpha)/K}(\alpha) x^{[K(\alpha):K]-1} + \dots + (-1)^{[K(\alpha):K]} N_{K(\alpha)/K}(\alpha)$$

then

$$f(x)^{[F:K(\alpha)]} = x^{[F:K]} - [F:K(\alpha)] \operatorname{Tr}_{K(\alpha)/K}(\alpha) x^{[F:K]-1} + \dots + (-1)^{[F:K]} N_{K(\alpha)/K}(\alpha)^{[F:K]}$$

(vii) Every embedding $\sigma: F \rightarrow \bar{K}$ extends to an automorphism of $\bar{K}$. The distinct embeddings are given by any complete set of coset representatives of $\operatorname{Aut}_F(\bar{K})$ in $\operatorname{Aut}_K(\bar{K})$

Theorem: Let F/K be a finite cyclic (Galois) extension

$$\operatorname{Gal}(F/K) = \langle \sigma \rangle$$

(i) If $\alpha \in F$, $\operatorname{Tr}_{F/K}(\alpha) = 0 \Leftrightarrow \alpha = \beta - \sigma(\beta)$ for some $\beta \in F$.

(ii) If $\alpha \in F$, $N_{F/K}(\alpha) = 1 \Leftrightarrow \alpha = \frac{\beta}{\sigma(\beta)}$ for some $\beta \in F$

Proof: In one direction,

$$\begin{aligned}\text{Tr}(\beta - \sigma(\beta)) &= \sum_{k=0}^{n-1} \sigma^k(\beta - \sigma(\beta)) = \sum_{k=0}^{n-1} \sigma^k(\beta) - \sigma^{k+1}(\beta) \\ &= \beta - \sigma^n(\beta) = 0\end{aligned}$$

Same for norm.

Claim: Let $\sigma_1, \dots, \sigma_n$ be distinct automorphisms of a field E . Then $\sigma_1, \dots, \sigma_n$ are linearly independent.

If not, assume there exists some $\sum_{k=1}^n a_k \sigma_k = 0$ with a minimal non-zero coefficients. Construct another linear combination with one fewer non-zero coefficient.

Given the claim, assume $\text{Tr}_{F/K}(\alpha) = 0$. We show that we can solve $\text{Tr}_{F/K}(w) = 1$. Since $\sum_{k=1}^n \sigma^k \neq 0$, so there exists $u \in F$ with $\text{Tr}_{F/K}(u) = \sum_{k=1}^n \sigma^k(u) \neq 0$. If $w = \text{Tr}_{F/K}^{-1}(u)u$

$$\text{Tr}_{F/K}(w) = \text{Tr}_{F/K}(\text{Tr}_{F/K}^{-1}(u))(u) = 1$$

$$\begin{aligned}\text{Let } \beta &= \alpha w + (\alpha + \sigma(\alpha))\sigma(w) + (\alpha + \sigma(\alpha) + \sigma^2(\alpha))\sigma^2(w) \\ &\quad + \dots + (\alpha + \sigma(\alpha) + \dots + \sigma^{n-2}(\alpha))\sigma^{n-2}(w)\end{aligned}$$

Now

$$\begin{aligned}\beta - \sigma(\beta) &= \alpha w + (\alpha + \sigma(\alpha))\sigma(w) + \dots + (\alpha + \sigma(\alpha) + \dots + \sigma^{n-2}(\alpha))\sigma^{n-2}(w) \\ &\quad - \sigma(\alpha)\sigma(w) - \sigma(\downarrow)\sigma^2(w) + \dots + \sigma(\downarrow)\sigma^{n-1}(w) \\ &= \alpha w + \alpha\sigma(w) + \alpha\sigma^2(w) + \dots + \alpha\sigma^{n-1}(w) - (\downarrow) \\ &= \alpha \text{Tr}_{F/K}(\alpha) - \text{Tr}_{F/K}(\alpha)\sigma^{n-1}(w) \\ &= \alpha\end{aligned}$$

Cyclic Extensions : 2 Cases

(i) $\text{char}(K) = 0$ or F/K in char p with $p \nmid [F:K]$

(ii) $\text{char}(K) = p$ with $[F:K] = p$.

Theorem: (Artin-Schrier) Let $\text{char}(K) = p$. The F/K is a cyclic extension of degree p if and only if F is the splitting field of some irreducible polynomial $x^p - x - A$ in $K[x]$.

Proof: If $f(x) = x^p - x - A$ is irreducible and $\alpha \in \bar{K}$ is a root. Then for any $i \in \mathbb{F}_p \subseteq K$

$$(\alpha + i)^p - (\alpha + i) - A = \alpha^p + i^p - \alpha - i - A = i^p - i = 0$$

So $x^p - x - A$ has p distinct roots in $K(\alpha)$. Thus $K(\alpha)$ is a sep.

splitting field so $F = K(\alpha)$ is a Galois extension of K . Each σ in $\text{Gal}(F/K)$ is determined by $\sigma(\alpha)$ and $\sigma(\alpha) = \alpha + i$ for some $i \in \mathbb{F}_p$, so $\sigma \mapsto \sigma(\alpha) - \alpha$ gives a map an isomorphism

$$\text{Gal}(F/K) \rightarrow \mathbb{Z}_p.$$

Now assume $[F:K] = p$, F/K is Galois and

$\text{Gal}(F/K) = \langle \sigma \rangle$. By the claim

$$\text{Tr}(1) = [F:K] \cdot 1 = p = 0.$$

So $\exists \beta \in F$ with $1 = \beta - \sigma(\beta)$. Put $\alpha = -\beta$, $\sigma(\alpha) = \alpha + 1$

$$\text{Note } \sigma(\alpha^p - \alpha) = \sigma^p(\alpha) - \sigma(\alpha) = (\alpha + 1)^p - (\alpha + 1) = \alpha^p - \alpha$$

So $\alpha^p - \alpha \in K$. Put $A = \alpha^p - \alpha$, so α is a root of $f(x)$

$f(x) = x^p - x - A = x^p - x - \alpha^p + \alpha = 0$, is irreducible because if

$\alpha \notin K$, $\sigma(\alpha) = \alpha + 1 \neq \alpha$, so $[K(\alpha):K] \neq 1$, but

$[K(\alpha):K] \mid [F:K] = p$, so $[K(\alpha):K] = p \Rightarrow F = K(\alpha)$

On the other hand $[K(\alpha):K] = \deg(\text{min poly})$, so $x^p - x - A$ is irreducible.